Name
Date
Course

Practice Beginning Algebra Test III (version 2)

Instructions

i) As always, write a title, name, date, and course.
ii) Number each problem, circle the number, and show work when necessary.
iii) Draw a rectangle around your answers.

Solve.

1. $4\sqrt{y}-6=-8$

2. $0=1{,}000x^3-40x$

3. $x^2-10x=11$

4. $x^3+2x^2-49x-98=0$

5. $-7=2-10\sqrt{x}$

6. $5\sqrt{2x+7}-12=-2$

⑦ $\sqrt{9x^2-35}=\sqrt{6x}$

⑧ $16b^2+2=7$

⑨ $8x^2=-10x$

⑩ $\dfrac{\left(\frac{x}{15}-2\right)^2}{4}+9=6$

11 $\frac{x}{4}+\frac{1}{6}=4+x$

12 $18x^9+50x^5=68x^7$

13 $2x+\frac{33}{x}=17$

14 $3=(4+x)^2+3$

⑮ $\frac{1}{y^3} = \frac{10}{y^4} - \frac{24}{y^5}$

⑯ $\frac{2}{3} + \frac{7}{9x} = -1 + \frac{1}{x}$

⑰ $x^3 + 8x^2 = 20x$

⑱ $6x^3 - 9x^2 = 3 - 2x$

19) $0 = x^3 - x^2 + x - 1$

20) $5 + 2\left(\frac{x}{6} + 2\right)^2 = 8$

21) $\frac{3x-3}{x^2-9x+8} - \frac{6}{x-8} = \frac{1}{x-1}$

22) $\frac{8}{x(x-2)} - \frac{1}{2x} = \frac{4}{x-2}$

(23) $$\frac{2}{x-5} = \frac{26}{x^2+x-30} + \frac{x-7}{x+6}$$

(24) $\sqrt{x+3} = \sqrt{x} + 2$

(25) $y+8 = \sqrt{8+y}$

Practice Beginning Algebra Test III (version 2) Answers

1. $\boxed{\text{No Solution}}$
2. $\boxed{x=0}$, $\boxed{x=\frac{1}{5}}$, $\boxed{x=-\frac{1}{5}}$
3. $\boxed{x=11}$, $\boxed{x=-1}$
4. $\boxed{x=-7}$, $\boxed{x=7}$, $\boxed{x=-2}$
5. $\boxed{x=81/100}$
6. $\boxed{x=-3/2}$
7. $\boxed{x=\frac{7}{3}}$, $x\neq-\frac{5}{3}$
8. $\boxed{b=\pm\sqrt{5}/4}$
9. $\boxed{x=-5/4}$, $\boxed{x=0}$
10. $\boxed{\text{No Solution}}$
11. $\boxed{x=-46/9}$
12. $\boxed{x=0}$, $\boxed{x=\frac{5}{3}}$, $\boxed{x=-\frac{5}{3}}$, $\boxed{x=1}$, $\boxed{x=-1}$
13. $\boxed{x=3}$, $\boxed{x=11/2}$
14. $\boxed{x=-4}$
15. $\boxed{x=6}$, $\boxed{x=4}$
16. $\boxed{x=2/15}$
17. $\boxed{x=0}$, $\boxed{x=2}$, $\boxed{x=-10}$
18. $\boxed{x=3/2}$
19. $\boxed{x=1}$
20. $\boxed{x=\pm3\sqrt{6}-12}$
21. $\boxed{x=11/4}$
22. $\boxed{\text{No Solution}}$
23. $\boxed{x=7}$
24. $\boxed{\text{No Solution}}$
25. $\boxed{y=-7}$, $\boxed{y=-8}$